

Telecom Cyber Security Project Monitoring

Hybrid Automation Framework: Python Exception Engine + Zapier Orchestration

Author: Business Analysis Team

Focus Area: Portfolio Risk & Cost Control

Status: Active Execution

Executive Summary

This report presents the operational framework for monitoring 10 critical Telecom Cyber Security projects. To ensure timely governance without overwhelming project leadership, a **Hybrid Automation Model** has been designed. Python serves as the analytical core, processing project health metrics, variance percentages, and schedule progress. Identified exceptions are seamlessly routed into **Zapier** to orchestrate targeted management alerts, email notifications, and automated status dashboard updates.

KEY PORTFOLIO PERFORMANCE INDICATORS

\$2.83M TOTAL APPROVED BUDGET	\$2.82M TOTAL ACTUAL SPEND	-\$10,000 PORTFOLIO COST VARIANCE	60% PROJECTS NEEDING ACTION
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PROJECT MONITORING MASTER DATASET

ID	PROJECT NAME	BUDGET (\$)	ACTUAL (\$)	ACT %	PLAN %	STATUS
101	NextGen Firewall Upgrade	\$250,000	\$265,000	85.0%	100%	Delay
102	SOC Implementation	\$600,000	\$570,000	76.25%	100%	Risk
103	ISO 27001 Compliance Audit	\$180,000	\$165,000	91.25%	100%	On Track
104	DDoS Protection Enhancement	\$320,000	\$350,000	65.0%	100%	Delay
105	Identity & Access Management	\$280,000	\$260,000	82.5%	100%	On Track
106	SIEM Migration Project	\$450,000	\$430,000	63.75%	100%	Risk
107	Cloud Security Assessment	\$150,000	\$175,000	60.0%	100%	Delay
108	Vulnerability Mgmt Upgrade	\$210,000	\$200,000	86.25%	100%	On Track
109	Privileged Access Management	\$300,000	\$290,000	66.25%	100%	Risk
110	Security Awareness Programme	\$90,000	\$85,000	92.5%	100%	On Track

HYBRID ARCHITECTURE STRATEGY OVERVIEW

The solution enforces a clear business analysis principle: **Data** → **Analysis** → **Automation** → **Action**. Raw database values are first evaluated using Python scripts to identify subtle performance gaps (>10% lag) and budget overruns

PYTHON ANALYTICAL ENGINE IMPLEMENTATION

The code snippet below represents the analytical component that acts as the intelligence layer prior to Zapier trigger events:

```
import pandas as pd

# Telecom Cyber Security Project Monitoring Automation Data
projects = [
    [101, "NextGen Firewall Upgrade", 250000, 265000, 85, 100, "Delay"],
    [102, "SOC Implementation", 600000, 570000, 76.25, 100, "Risk"],
    [103, "ISO 27001 Compliance Audit", 180000, 165000, 91.25, 100, "On Track"],
    [104, "DDoS Protection Enhancement", 320000, 350000, 65, 100, "Delay"],
    [105, "Identity & Access Management Upgrade", 280000, 260000, 82.5, 100, "On Track"],
    [106, "SIEM Migration Project", 450000, 430000, 63.75, 100, "Risk"],
    [107, "Cloud Security Assessment", 150000, 175000, 60, 100, "Delay"],
    [108, "Vulnerability Management Upgrade", 210000, 200000, 86.25, 100, "On Track"],
    [109, "Privileged Access Management", 300000, 290000, 66.25, 100, "Risk"],
    [110, "Security Awareness Programme", 90000, 85000, 92.5, 100, "On Track"]
]

columns = ["Project_ID", "Project_Name", "Budget_USD", "Actual_Cost_USD",
           "Actual_Percent", "Planned_Percent", "Status"]
df = pd.DataFrame(projects, columns=columns)

# 1. Performance Gap & Financial Variance
df["Performance_Gap"] = df["Actual_Percent"] - df["Planned_Percent"]
df["Cost_Variance"] = df["Actual_Cost_USD"] - df["Budget_USD"]
df["Cost_Variance_Percent"] = (df["Cost_Variance"] / df["Budget_USD"] * 100).round(2)

# 2. Identify Projects Requiring Action
df["Action_Required"] = "No"
df.loc[(df["Performance_Gap"] < -10) | (df["Cost_Variance"] > 0) |
       (df["Status"].isin(["Delay", "Risk"])), "Action_Required"] = "Yes"
```

CALCULATED VARIANCE & ANALYTICAL OUTPUT

Execution of the analysis flags projects requiring executive visibility due to schedule delay, financial overruns, or risk designations:

PROJECT NAME	PERF. GAP	COST VAR (\$)	COST VAR %	STATUS	ACTION FLAG
NextGen Firewall Upgrade	-15.0%	+\$15,000	+6.00%	Delay	YES
SOC Implementation	-23.75%	-\$30,000	-5.00%	Risk	YES
ISO 27001 Compliance Audit	-8.75%	-\$15,000	-8.33%	On Track	No
DDoS Protection Enhancement	-35.0%	+\$30,000	+9.38%	Delay	YES
SIEM Migration Project	-36.25%	-\$20,000	-4.44%	Risk	YES
Cloud Security Assessment	-40.0%	+\$25,000	+16.67%	Delay	YES
Privileged Access Mgmt	-33.75%	-\$10,000	-3.33%	Risk	YES

ZAPIER INTEGRATION ARCHITECTURE

The business analysis framework enforces four discrete structural phases to transform raw telemetry into high-value managerial action:

